Consider the Lévy density

$$k(x) = C\frac{\exp\left(\left(G-M\right)x/2 - (G+M)|x|/2\right)}{|x|^{1+Y}(1+|x|)^W}$$

The purpose is to have two parameters working away from zero, on the positive side M, W that on a measure change can allow P to be above Q for low x values with the reverse holding for large x values and C, Y not being allowed to change to have the Hellinger condition for equivalent measures.

For the log characteristic function we have to evaluate

$$\begin{aligned}
&\int_{-\infty}^{\infty}\left(e^{iux}-1\right)k(x)dx\\
=&\int_{-\infty}^{\infty}\left(e^{iux}-1\right)C\frac{\exp\left(\left(G-M\right)x/2-(G+M)|x|/2\right)}{|x|^{1+Y}(1+|x|)^W}dx\\
=&\int_{-\infty}^{0}\left(e^{iux}-1\right)C\frac{\exp(-G|x|)}{|x|^{1+Y}(1+|x|)^W}dx+\int_{0}^{\infty}\left(e^{iux}-1\right)C\frac{\exp\left(-Mx\right)}{|x|^{1+Y}(1+|x|)^W}dx\\
=&\int_{0}^{\infty}\left(e^{-iux}-1\right)C\frac{\exp(-Gx)}{x^{1+Y}(1+x)^W}dx+\int_{0}^{\infty}\left(e^{iux}-1\right)C\frac{\exp\left(-Mx\right)}{|x|^{1+Y}(1+|x|)^W}dx\\
=&C\left[\int_{0}^{\infty}\frac{\exp\left(-(G+iu)x\right)}{x^{1+Y}(1+x)^W}dx-\int_{0}^{\infty}\frac{\exp(-Gx)}{x^{1+Y}(1+x)^W}+\int_{0}^{\infty}\frac{\exp\left(-(M-iu)x\right)}{x^{1+Y}(1+x)^W}dx-\int_{0}^{\infty}\frac{\exp\left(-Mx\right)}{x^{1+Y}(1+x)^W}dx\right.
\end{aligned}$$

We just need to evaluate

$$\begin{aligned}
\Psi(z) &= \int_{0}^{\infty}\frac{\exp(-zx)}{x^{1+Y}(1+x)^W}dx\\
&= z^Y\int_{0}^{\infty}\frac{u^{-Y-1}e^{-u}}{(1+\frac{u}{z})^W}du\\
&= \frac{z^Y}{\Gamma(W)}\int_{0}^{\infty}u^{-Y-1}e^{-u}du\int_{0}^{\infty}a^{W-1}e^{-(1+\frac{u}{z})a}da\\
&= \frac{z^Y}{\Gamma(W)}\int_{0}^{\infty}e^{-a}a^{W-1}\int_{0}^{\infty}u^{-Y-1}e^{-(1+\frac{a}{z})u}du\\
&= \frac{z^Y}{\Gamma(W)}\int_{0}^{\infty}e^{-a}a^{W-1}\int_{0}^{\infty}e^{-w}(\frac{z}{z+a})^{-Y}w^{-Y-1}dw\\
&= \frac{\Gamma(-Y)}{\Gamma(W)}\int_{0}^{\infty}e^{-a}a^{W-1}(z+a)^Y da\\
&= \Gamma(-Y)z^{K+Y}U(K,K+Y,z)
\end{aligned}$$

For $Y>0$ this is a regularized value obtained by analytic continuation and is not equal to the divergent integral. They may be used with the correction terms that come in with characteristic exponent which is a convergent integral.

So the characteristic exponent is

$$\begin{aligned}
&C_n\Gamma(-Y_n)\left((G+iu)^{W_n+Y_n}U(W_n,W_n+Y_n,G+iu)-G^{W_n+Y_n}U(W_n,K_n+Y_n,G)\right)\\
&+C_p\Gamma(-Y_p)\left((M-iu)^{W_p+Y_p}U(W_p,W_p+Y_p,M-iu)-M^{W_p+Y_p}U(W_p,W_p+Y_p,M)\right)
\end{aligned}$$

where $U(a, b, z)$ is the *kummerU* function.

So we have a new model with a closed form using special functions. The model may be estimated on time series data where it will have a drift which is then risk-neutralized by choosing minimal norm values for G, M, W_n, W_p that meet the martingale condition. We may also obtain a Q measure by calibrating G, M, W_n, W_p to option data. Just an exponential tilt is not rich enough for option data. Now we can use power tilts as well as exponential tilts.

I am opening a section in FMF titled "Rapid Communications in Mathematical Finance", that will take 5 page papers. I will send it there.